

Multivariable Chapter 1

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1 Problems

P 1.7.10. I claim that if the derivative exists, it is unique. Suppose

$$\lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - D \cdot h) = 0$$

and

$$\lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - E \cdot h) = 0.$$

Then

$$\lim_{h \rightarrow 0} \frac{1}{|h|} ((D - E) \cdot h) = \lim_{h \rightarrow 0} (D - E) \cdot u = 0,$$

where u is a unit vector. We can pick u to be parallel to any basis direction, so $D = E$.

Now I claim for a linear transformation $g : v \mapsto Mv$ at any a , $D(g)(a) = M$. We have

$$\lim_{h \rightarrow 0} \frac{1}{|h|} (M(a+h) - Ma - Mh) = 0,$$

so indeed $D(g)(a) = M$.

Suppose f is affine. Then $g(v) = f(v) - f(0) = Mv$ is a linear transformation. $f(a+h) - f(0) = M(a+h)$, $f(a) - f(0) = Ma$, $f(h) - f(0) = Mh$, so

$$f(a+h) - f(a) = M(a+h) - Ma = Mh = f(h) - f(0).$$

We write:

$$\lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - Mh) = \lim_{h \rightarrow 0} \frac{1}{|h|} (f(h) - f(0) - Mh) = 0,$$

so $M = D(f)(a)$ for f at all a .

Finally,

$$f(a+v) - f(a) = f(v) - f(0) = Mv = D(f)(a)v,$$

showing the result.

P 1.7.15. (a) A map $F : \text{Mat}(n \times m) \rightarrow \text{Mat}(k \times l)$ is said to be differentiable at A if there exists a linear transformation $L : \text{Mat}(n \times m) \rightarrow \text{Mat}(k \times l)$ for which

$$\lim_{H \rightarrow 0} \frac{1}{|H|} (F(A + H) - F(A) - L(H)) = 0.$$

(b)

$$\begin{aligned} & (A + H)(A + H)^\top - AA^\top \\ &= (A + H)(A^\top + H^\top) - AA^\top \\ &= AH^\top + HA^\top + HH^\top. \end{aligned}$$

I claim $L(H) = AH^\top + HA^\top$ is a linear transformation.

$$\begin{aligned} & L(H_1 + H_2) \\ &= A(H_1 + H_2)^\top + (H_1 + H_2)A^\top \\ &= AH_1^\top + AH_2^\top + H_1A^\top + H_2A^\top \\ &= (AH_1^\top + H_1A^\top) + (AH_2^\top + H_2A^\top) \\ &= L(H_1) + L(H_2). \end{aligned}$$

And for $c \in \mathbb{R}$:

$$L(cH) = A(cH)^\top + (cH)A^\top = c(AH^\top + HA^\top) = cL(H).$$

Hence, L is indeed a linear transformation. Now,

$$\frac{1}{|H|} (|HH^\top|) \leq \frac{1}{|H|} |H| |H^\top| = |H|,$$

which tends to 0 as $H \rightarrow 0$. $D(F)(H) = AH^\top + HA^\top$.

P 1.7.21.

$$\begin{aligned} & (1 + h_{1,1})(1 + h_{2,2}) - h_{1,2}h_{2,1} - 1 - (h_{1,1} + h_{2,2}) \\ &= h_{1,1}h_{2,2} - h_{1,2}h_{2,1} \\ &\leq \frac{1}{2}(h_{1,1}^2 + h_{1,2}^2 + h_{2,1}^2 + h_{2,2}^2) \\ &= |H|^2/2. \end{aligned}$$

Hence, $\lim_{H \rightarrow 0} |H|^2 / (2|H|) = 0$.

P 1.7.11. (a)

$$J(f) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \cos xy & x \cos xy \end{pmatrix}.$$

(b)

$$J(f) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2xe^{x^2+y^3} & 3y^2e^{x^2+y^3} \end{pmatrix}.$$

(c)

$$J(f) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y & x \\ 1 & 1 \end{pmatrix}$$

.

(d)

$$J(f) \begin{pmatrix} r \\ \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}.$$

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$$\lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - [0] \cdot h) = 0.$$

2.

$$\lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - Mh) = \lim_{h \rightarrow 0} \frac{1}{|h|} (M(a+h) - Ma - Mh) = 0.$$

4.

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) + g(a+h) - f(a) - g(a) - Df(a)(h) - Dg(a)(h)) \\ &= \lim_{h \rightarrow 0} \frac{1}{|h|} (f(a+h) - f(a) - Df(a)(h)) + \lim_{h \rightarrow 0} \frac{1}{|h|} (g(a+h) - g(a) - Dg(a)(h)) \\ &= 0 + 0 \\ &= 0, \end{aligned}$$

so $D(f+g)(a) = Df(a) + Dg(a)$.

5.